

RITIK AGARWAL

Full-Stack Developer | B.Tech CSE, VIT Bhopal | India

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SUMMARY

Full-stack developer and 3rd-year CS student who builds and deploys complete products — frontend, APIs, database and hosting — across TypeScript/Node and Java/Spring Boot. As a full-stack intern at Labmentix I have shipped four products, and I write up the decisions behind my work.

EDUCATION

Vellore Institute of Technology, Bhopal — B.Tech in Computer Science & Engineering Sep 2024 – 2028
3rd Year · Expected Graduation: 2028

Relevant coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks, System Design (Self-Study)

EXPERIENCE

Full-Stack Web Development Intern — Labmentix Jun 2026 – Present

Code: [SmartERP](#) | [PDF Sign](#) | [CloudVault](#)

- SmartERP — multi-tenant billing, inventory and accounting ERP (Next.js, Express, Prisma, PostgreSQL): company-scoped data isolation, double-entry validation (debits = credits) inside database transactions, GST invoices with PDF export.
- PDF Sign — document-signing app (React, Express, pdf-lib): drag-and-drop signature placement, per-signer signing links secured by random 256-bit tokens, signed-PDF generation, email notifications and an audit trail.
- CloudVault — cloud file storage in Java Spring Boot + React: direct-to-storage uploads via signed URLs, viewer/editor sharing, password-protected expiring public links, and hashed refresh tokens in HttpOnly cookies.

Senior Marketing Manager — AIESEC in Bhopal Feb 2026 – Present

- Promoted from Junior to Senior Marketing Manager within six months (Feb → Aug 2026); serve on the Governing Board of the marketing department.

PROJECTS

TripSync — AI Trip Planner with Real-World Constraints [Live](#) | [Code](#)

Stack: Next.js, TypeScript, PostgreSQL, Neon, Drizzle ORM, Better Auth, Groq / Gemini APIs, Tailwind CSS

- Generates day-by-day itineraries that respect opening hours, meal times, travel time between stops and a fixed budget; output is validated as structured JSON and stored as relational rows.
- Routed generation through an LLM provider chain (Groq, then Gemini) ordered by an environment variable, so a provider outage is a restart rather than a redeploy.
- Fixed an "infinite recursion detected in policy" failure between two row-level-security policies by moving access control into one owner/editor/viewer check; later migrated from Supabase to Neon Postgres with Drizzle ORM.

Voxora — Multilingual Text-to-Speech that Translates First (Labmentix internship) [Live](#) | [Code](#)

Stack: React, Vite, Node.js, Express, PostgreSQL, Neon, JWT Auth, Cloudflare R2

- Built a translate-then-synthesise pipeline so the chosen language is the one you hear — 30 neural voices across 15 languages including Hindi, Bengali and Tamil.
- Diagnosed a production-only failure: server-side translation was IP rate-limited (HTTP 429/403); moved translation to the browser so each visitor uses their own IP, keeping a server fallback.
- Moved generated audio from an ephemeral filesystem to Cloudflare R2 behind a storage interface; verified by deleting local files and confirming playback still returned HTTP 200.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Java, Python, C/C++, SQL

Frontend: React, Next.js, Vite, Tailwind CSS

Backend: Node.js, Express, Spring Boot, REST APIs, JWT Auth, Better Auth

Databases: PostgreSQL, MongoDB, MySQL, Redis, Prisma, Drizzle ORM

Cloud & AI: Groq / Gemini APIs, Supabase, Neon, Vercel, Render, Cloudflare R2, Git & GitHub

CERTIFICATIONS

The Bits and Bytes of Computer Networking — Google (Coursera) Jul 2026